

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1960

Simon Street 138 kV
July 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1960. The Interconnection Customer proposes a 187.5 MW solar photovoltaic facility interconnecting at the Simon Street 138 kV substation in the Pennsylvania area. The request was received on December 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 548 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,255,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Simon St / Slater 138.0 kV	Mighowrd Substation / Simon St 138.0 kV line	152.2	287	100.0
Foster Wheeler Mt Carmel Cogen / Mighowrd Substation 138.0 kV	Slater / Simon St 138.0 kV line	135.0	315	100.0
Mighowrd / Simon St 138.0 kV Line	Simon St / Slater 138.0 kV L	132.9	329	100.0
Frackville / North Market 138.0 kV Line	Mighowrd / Simon St 138.0 kV	128.5	267	58.9
N Market / Roseville 138.0 kV Line	Mighowrd / Simon Street 138.0 kV L	127.1	277	58.9
Roseville / Dug Gap 138.0 kV	Mighowrd / Simon Street 138.0 kV	124.8	293	58.9
Dug Gap / George 138.0 kV	Mighowrd / Simon St 138.0 kV	123.6	303	58.9
Slater / George 138.0 kV	Simon St / Mighowrd 138.0 kV	122.4	314	58.9
Sulphur / Foster Wheeler Mt Carmel Cogen 138.0 kV	Foster Wheeler Mt Carmel Cogen / Westwood Generation Llc Substation 345.0 kV	106.8	223	19.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Slater / Simon St 138.0 kV	Rebuild line with higher-capacity conductor	\$2,131,000	42% (\$900,000)
Foster Wheeler Mt Carmel Cogen / Mighowrd Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$4,117,000	42% (\$1,739,000)
Mighoward / Simon St 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,169,000	42% (\$2,605,000)
Frackville / N Market 138.0 kV Line	Rebuild line with higher-capacity conductor	\$15,055,000	42% (\$6,358,000)
N Market / Roseville 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,738,000	42% (\$2,001,000)
Roseville / Dug Gap 138.0 kV	Rebuild line with higher-capacity conductor	\$2,561,000	42% (\$1,082,000)
Dug Gap / George 138.0 kV	Rebuild line with higher-capacity conductor	\$3,456,000	42% (\$1,459,000)
Slater / George 138.0 kV	Rebuild line with higher-capacity conductor	\$2,988,000	42% (\$1,262,000)
Sulphur / Foster Wheeler Mt Carmel Cogen 138.0 kV	Reconductor line	\$2,147,000	40% (\$849,000)
	Total allocated to project		\$18,255,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.